

Victor Ndunda

AI Engineer · Founder (Busara AI & KilimoPRO) · Researcher

Nairobi, Kenya · +254 724 346 971 · mututandunda@gmail.com
victorndunda.com · github.com/gadda00 · linkedin.com/in/victor-ndunda

PROFESSIONAL SUMMARY

AI engineer and founder building production multi-agent intelligence systems and agricultural AI for African markets. Founder of Busara AI — a 50-agent data intelligence platform running a 7-stage parallel DAG pipeline with real statistical algorithms (Holt-Winters, OLS, K-Means++, Granger causality, GARCH). Also building KilimoPRO, an offline-first Flutter app for African farmers with on-device crop disease detection (TensorFlow Lite), market price forecasting, and a multi-LLM advisory council integrating 22+ data sources. 5+ years shipping TypeScript, Python, Flutter, and ML systems that serve real users.

EXPERIENCE

Founder & Lead AI Engineer — Busara AI

2023 — Present

busaraai.com · Nairobi, Kenya

- Designed and shipped a 50-agent multi-agent platform with a 7-stage parallel DAG pipeline (Ingest → Engineer → Detect → Forecast → Infer → Report) with circuit breakers and smart caching — serving live users at busaraai.com.
- Implemented real statistical algorithms from scratch: Holt-Winters forecasting, Z-score/IQR/EWMA anomaly ensembles, OLS regression, Granger causality, K-Means++ clustering.
- Integrated GLM-4.6, GPT, Claude, and Gemini into a multi-LLM council with streaming responses and citation-grounded RAG.
- Cut average analysis latency by ~40% via smart caching and parallel DAG execution.
- Stack: TypeScript, Next.js 15, React 19, PostgreSQL, Redis, Docker, Kubernetes, PWA.

Founder & Lead Engineer — KilimoPRO

2022 — Present

kilimo.pro · github.com/gadda00/kilimopro · Kenya

- Built an offline-first Flutter app for African farmers with on-device crop disease detection via TensorFlow Lite — runs on \$80 Android phones.
- Engineered market price forecasting using GARCH/EWMA/VaR, and a multi-LLM advisory council covering agronomy, markets, climate, and finance.
- Integrated 22+ data sources: KilimoSTAT, FAOSTAT, Google Earth Engine, KALRO, open weather APIs.
- Aligned with Kenya's NASIP 2026-2030 agricultural strategy; supports SMS/USSD channels for low-end phone users.
- Open-sourced on GitHub; designed for scale across East African smallholder farmers.

AI / Data Engineer — Independent Consulting

2019 — 2022

Remote · Nairobi

- Delivered financial forecasting systems (ARIMA, exponential smoothing, backtesting) for fintech clients — models in production for 12+ months.
- Built fraud-detection pipelines combining rule-based scoring with ML anomaly detection.
- Shipped customer-segmentation engines (K-Means, DBSCAN) for marketing analytics teams.
- Proved the multi-agent DAG architecture with IntelliFlow v2 — a 12-agent Python system that became the foundation of Busara AI.

EDUCATION

University of Embu — B.Sc. in Computer Science

2014 — 2018

Embu, Kenya

- Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Statistics, Software Engineering, Distributed Systems.
- Final-year project: an early crop advisory system that seeded the KilimoPRO concept.
- Academic achievement: Top-quartile standing in the School of Computing.

TECHNICAL SKILLS

Languages	TypeScript, Python, Dart, SQL, JavaScript
AI / ML	Multi-agent orchestration, RAG, LLM integration (GLM-4.6, GPT, Claude, Gemini), TensorFlow Lite, scikit-learn
Statistics	Holt-Winters, ARIMA, GARCH, OLS regression, Granger causality, K-Means++, DBSCAN, EWMA
Frontend	Next.js 15, React 19, Flutter, PWA, Tailwind CSS, accessibility (WCAG)
Backend	Node.js, FastAPI, PostgreSQL, Prisma, Supabase, Redis, REST + WebSocket
Infra / DevOps	Docker, Kubernetes, GitHub Actions, CI/CD, monitoring, circuit breakers
Data Sources	KilimoSTAT, FAOSTAT, Google Earth Engine, KALRO, open weather & market APIs
Payments / Locale	M-Pesa, Flutterwave, Stripe, Swahili/English i18n, SMS/USSD integration

SELECTED PROJECTS

Busara AI — 50-agent data intelligence platform	<i>Live · busaraai.com</i>
Production multi-agent system with a 7-stage parallel DAG pipeline. Real statistical math, RAG with citations, streaming responses. GLM-4.6-powered.	
KilimoPRO — AI for African farmers	<i>Open Source · kilimo.pro</i>
Flutter + TensorFlow Lite app. On-device crop disease detection, GARCH market forecasting, multi-LLM advisory council, 22+ data sources, SMS/USSD fallback.	
IntelliFlow v2 — 12-agent Python DAG system	<i>github.com/gadda00/IntelliFlow</i>
The architectural prototype that became Busara AI. Proved the multi-agent DAG pattern end-to-end.	

OPEN SOURCE & WRITING

- Open-source maintainer — github.com/gadda00 (KilimoPRO, IntelliFlow, Financial Forecasting, Fraud Detection, Customer Segmentation).
- Author of technical articles on multi-agent DAG architecture, real math vs vibes, multi-LLM councils, quant finance for agriculture, and 22 data sources — published at victorndunda.com/blog/.
- Author of 8 'How to Use AI' guides for SMBs — chatgpt for business, prompt engineering, AI tools, no-code chatbots, automation, AI for farmers, Swahili AI, RAG in one day.
- TikTok Developer Domain Verification — verified technical domain ownership.

CERTIFICATIONS & RECOGNITION

- PWA Build — offline-capable portfolio with service worker, manifest, and install prompt.
- WCAG-aware accessibility — skip-to-content, focus-visible, reduced-motion, ARIA labels.
- Aligned with Kenya's National Agricultural Information System (NASIP) 2026-2030.
- TikTok Developer Domain Verification — verified technical domain ownership.

LANGUAGES

English (Professional) · Swahili (Native) · Basic French